Range

1 Find the range of the following sets of scores:

a $10, 7, 2, 14, 13, 15, 11, 4$

b $15, -2, -8, 8, 15, 6, -16, 15$

c $-0.5537, 1.7444, -0.3381, 0.7200, -0.3381, 1.0435$

2 A group of students had a range in marks of 14 and the lowest score was 9. What was the highest score in the group?

3 The range of a set of scores is 8, and the highest score is 19. What is the lowest score in the set?

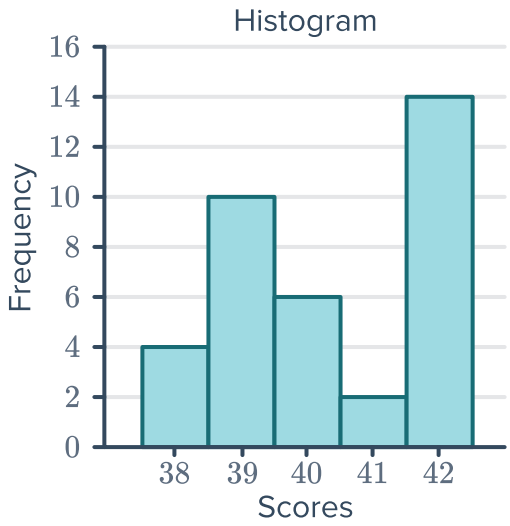
4 Consider the data provided in the table:

a Find the range of the scores.

b Find the mode.

Score	Frequency
68	16
69	41
70	30
71	31
72	49
73	29

5 Calculate the range for the data in the following histogram:



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Interquartile range

6 For each set of scores, find:

i The total number of scores

ii The median

iii The range

iv The first quartile

v The third quartile

vi The interquartile range

a $13, 15, 5, 16, 7, 20, 12$

b $-3, -3, 1, 9, 9, 6, -9$

c $33, 38, 50, 12, 33, 48, 41$

d $1.2, 2.9, 3.5, 1.6, 3.2, 4.8, 1.9$

e

	Leaf
2	2 5 6 7 9
3	0 0 5 6 8
4	0 0 1 8 9

Key: $5|2 = 52$

f

Score	Frequency
5	3
13	3
16	2
28	2
31	3
38	4
48	2

7 Consider the dot plot below:



a Find the total number of scores.

b Find the median.

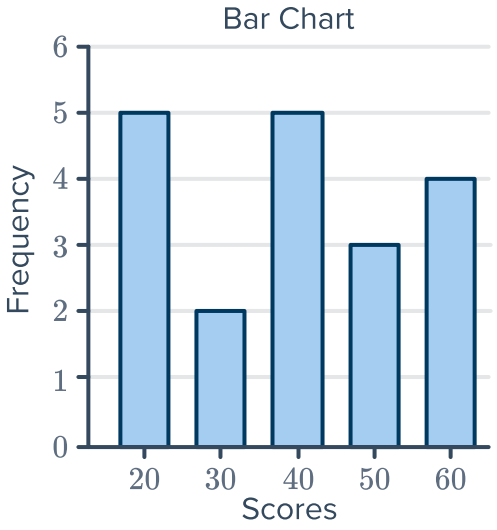
c Find the first quartile.

d Find the third quartile.

e Find the interquartile range.

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8 Consider the following set of scores displayed in the bar chart:



- **a** Construct a cumulative frequency table for this data.
- **b** Find the median score.
- **c** Find the first quartile.
- **d** Find the third quartile.
- **e** Find the interquartile range.

Quantiles, percentiles and deciles

9 Consider the data set below:

$$9, 5, 6, 3, 9, 8, 4, 2, 3, 2$$

Calculate:

- **a** The mean, to two decimal places.
- **b** The median.
- **c** The 1st quartile.
- **d** The 3rd quartile.
- **e** The 2nd decile.
- **f** The 8th decile.
- **g** The 43rd percentile.
- **h** The 88th percentile.

10 Consider the data provided in the table:
Calculate the value of:

- **a** The 4th decile.
- **b** The 7th decile.
- **c** The the 15th percentile.
- **d** The the 63rd percentile.

Observation	Cumulative frequency
2	8
3	15
4	21
5	28
6	31
7	34

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11 Consider the data provided in the table:
Calculate the value of:

Observation	Cumulative frequency
$5 - 9$	5
$10 - 14$	14
$15 - 19$	22
$20 - 24$	25
$25 - 29$	29
$30 - 34$	33

a The 3rd decile.

b The 8th decile.

c The the 23rd percentile to one decimal place.

d The the 68th percentile to one decimal place.

12 Fred was told he was ranked in the 80th percentile for mathematics in an end of year exam. However, his teacher only posted a list of results, without names, as shown below:

$$21, 29, 31, 33, 34, 42, 46, 47, 49, 58, 61, 63, 65, 72, 76, 81, 84, 87, 91, 95$$

a If there are 20 people in Fred's class, what was his class ranking?

b What result did Fred receive in the test?

13 Consider the following data set containing 100 scores in a 10×10 grid:

20	31	37	43	47	54	68	78	84	93
23	31	38	43	47	54	68	78	84	93
23	31	40	44	48	57	70	78	87	94
25	31	40	45	48	57	71	79	87	94
26	32	40	45	49	61	73	80	89	94
27	33	41	45	51	63	73	81	90	96
28	35	41	45	51	64	73	81	90	97
28	36	42	46	51	66	75	82	90	98
29	36	42	46	53	67	75	83	92	99
30	37	43	46	53	68	76	84	92	99

a Find the 69th percentile.

b Find the 28th percentile.

c What is the percentile of 68?

d What is the percentile of 43?

e Find the 3rd quartile.

f Find the 2nd decile.

g Find the 7th decile.

14 Charlie is in a table tennis league. He is ranked third out of 30 people. Calculate in what percentile his team lies.

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15 Roald is in a volleyball competition. His team is ranked 46th out of 180 teams. Calculate in what percentile his team lies. Give your answer to the nearest whole number.

Standard deviation and variance

16 Use the statistics mode on the calculator to determine the standard deviation of the following sets of scores. Round your answer to two decimal places.

a $-17, 2, -6, 9, -17, -9, 3, 8, 5$

b $8, 20, 16, 9, 9, 15, 5, 17, 19, 6$

17 For the given stem and leaf plot, calculate the following to one decimal place:

a The mean

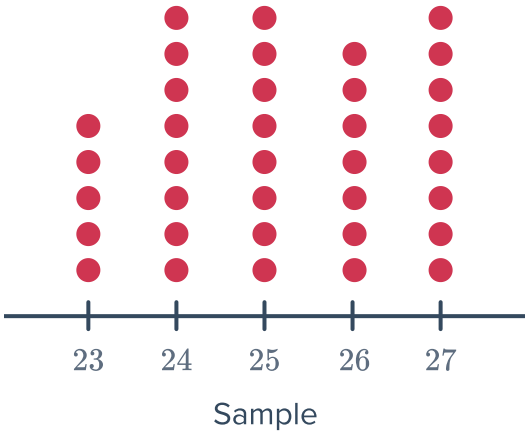
b The standard deviation

c The variance

	Leaf
1	3 3 9
2	3 8
3	4
4	4 7 9
5	5 8
6	8
7	5
8	3
9	7

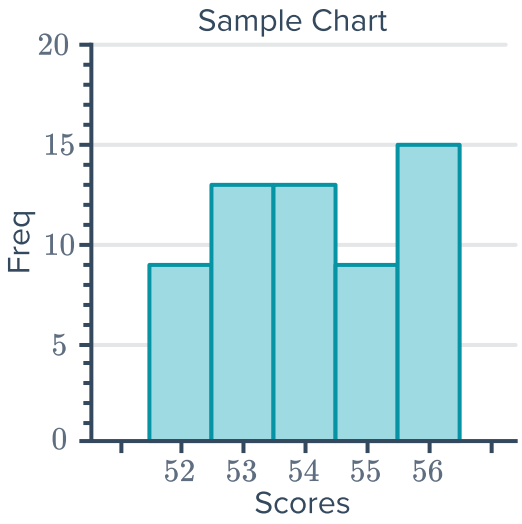
Key: $1|2 = 12$

18 Calculate the standard deviation for the data given in the following dot plot. Round your answer to two decimal places.



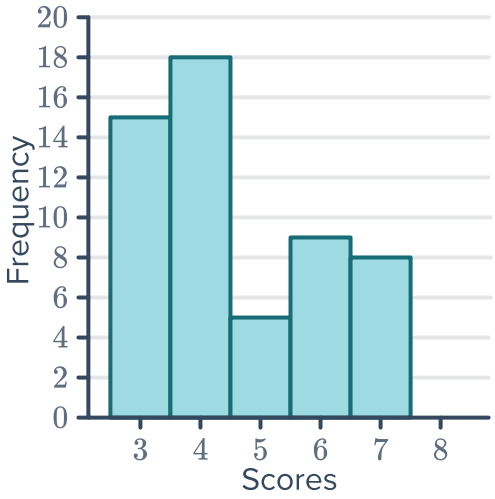
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19 Consider the given column graph:



- **a** Find the range of the data set.
- **b** Find the mean of the data set, correct to two decimal places.
- **c** Find the population standard deviation, correct to two decimal places.
- **d** Find the variance, correct to three significant figures.

20 Calculate the standard deviation for the following data represented by the frequency histogram. Round your answer to two decimal places.



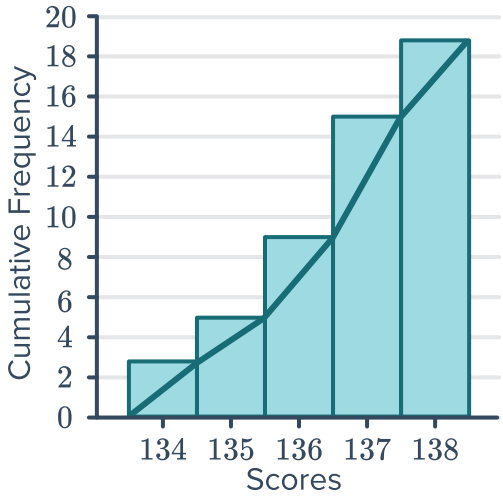
21 Consider the following table:

- **a** Complete the table.
- **b** Calculate an estimate for the mean. Round your answer to two decimal places.
- **c** Calculate an estimate for the standard deviation. Round your answer to two decimal places.
- **d** If we used the original ungrouped data to calculate standard deviation, would we expect it to have a higher or lower standard deviation? Explain your answer.

Class	Class centre (x)	f	fx
$1-9$		8	
$10-18$		6	
$19-27$		4	
$28-36$		6	
$37-45$		8	
Total			

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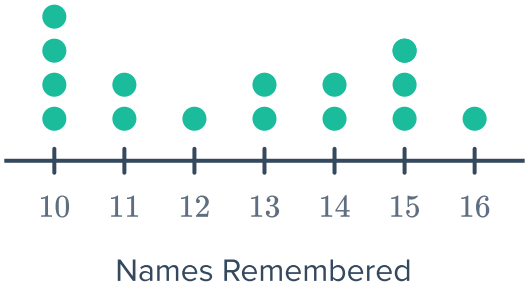
22 Consider the following cumulative frequency histogram:



- **a** Determine the range of scores.
- **b** Determine the mode.
- **c** Determine the median score.
- **d** Calculate the mean, correct to two decimal places.
- **e** Use your calculator to find the standard deviation. Round your answer to one decimal place.

Applications

23 In a study, a group of people were shown 30 names, and after 1 minute, they were asked to recite as many names by memory as possible. The results are presented in the dot plot:



- **a** What does each dot represent?
- **b** How many people took part in the study?
- **c** What is the largest number of names someone remembered?
- **d** What was the smallest number of names someone remembered?
- **e** What is the range?

24 A cyclist measured his heart rate immediately after finishing each event in which he competed. The results are recorded in the following stem plot:

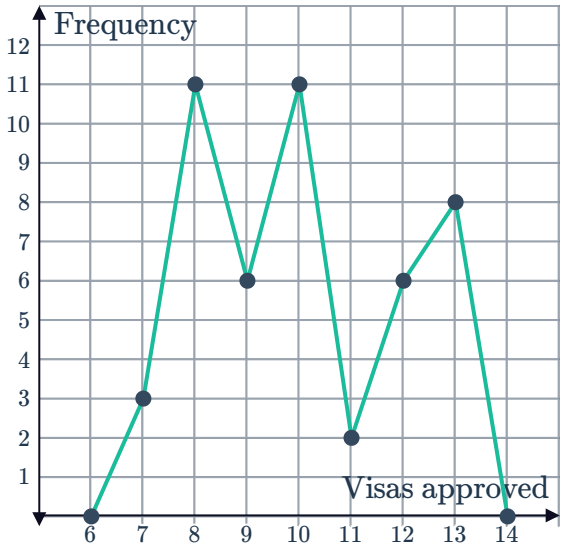
	Leaf
16	0 3 4 4 6 7 8
17	1 2 2 3 5 5 5 6
18	4 4 6 8
19	0 1 3 6

Key: $13|2 = 132$

- **a** What was the difference between his slowest and fastest heart rate?
- **b** For each extra heartbeat per minute, it takes 10 seconds longer for him to recover. How much longer would it take to recover when he finishes with the fastest heart rate than when he finishes with the slowest heart rate?

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25 The graph shows the number of visa applications approved by an embassy each day over a period of time:



- **a** Over how many days was the data recorded?
- **b** What was the median number of visas approved each day?
- **c** On what percentage of days were less than the median number of visas approved? Round your answer to two decimal places.
- **d** What is the lower quartile value?
- **e** What is the upper quartile value?
- **f** On how many days were between 8 and 12 visas (inclusive) approved?

26 The stem-and-leaf plot shows the results of a survey conducted on the price of concert tickets locally and the price of the same concerts at an international venue:

International		Local
9 9 9	**6**	8
8 5 5 5 3 0	**7**	5 6 6 9 9
8 4 3 2 1 0	**8**	2 2 6 6
5 3 2 0	**9**	0 0 1 4 5 6 8
5	**10**	0 3 5

Key: $9|6|8 = \$69 \text{ and } \$68$

- **a** Find the interquartile range for the international venue.
- **b** Find the interquartile range for the local venue.
- **c** At which venue is there the least spread in the middle 50% of prices?

27 The mean income of people in Country A is $\$19\,069$. This is the same as the mean income of people in Country B. The standard deviation of Country A is greater than the standard deviation of Country B. In which country is there likely to be the greatest difference between the incomes of the rich and poor?

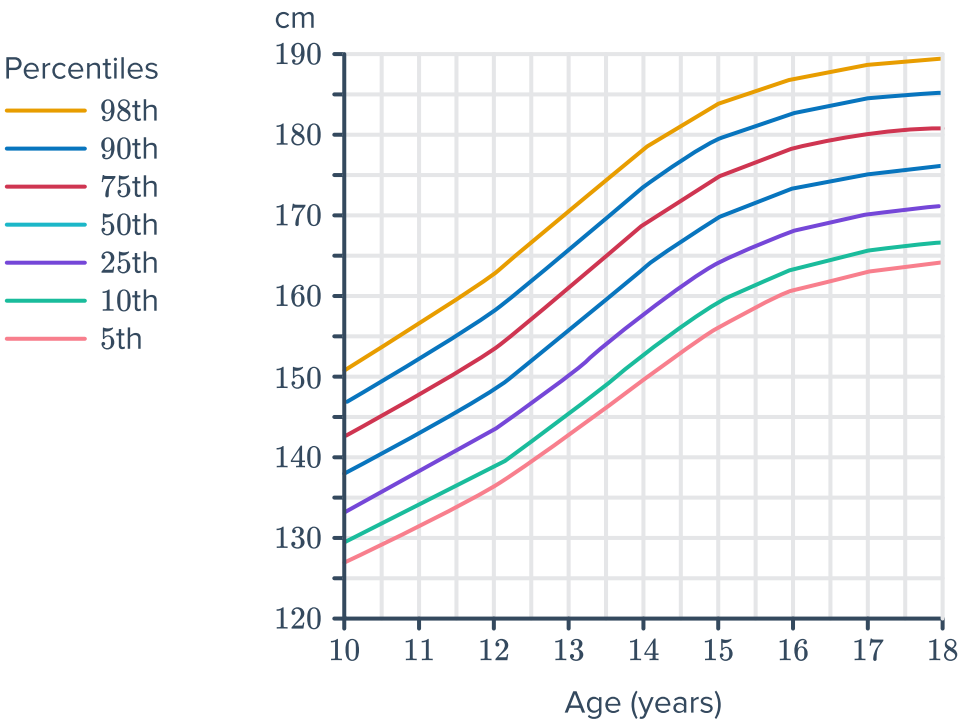
28 The scores of five diving attempts by a professional diver are recorded below:

$$5.6,\ 6.6,\ 6.3,\ 5.9,\ 6.4$$

- **a** Calculate the population standard deviation of the scores, correct to two decimal places.
- **b** On the sixth dive, the diver scores 8.8. What effect will this score have on the mean and standard deviation?
- **c** If each judge gave the same score for the 6th dive, state the standard deviation of the scores for this dive.

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29 Consider the following chart showing the range of heights for boys aged 10 to 18 years:



(Source: State Government of Victoria, Department of Education and Training)

a Lachlan is 15 and his height is at the 9th decile. Select the correct range of heights for Lachlan from the following:

A $182 - 185$ cm **B** $177 - 181$ cm **C** $167 - 172$ cm **D** $158 - 161$ cm

b Quentin is aged 14 and is 150 cm tall.

i What percentage of boys his age are taller than him?

ii If Quentin's height continues to follow this growth chart, how tall will he be when he is 18?

30 Meteorologists predicted a huge variation in temperatures throughout the month of April. The temperature each day for the first two weeks of April were recorded as follows:

$$16,\ 18,\ 20.5,\ 21,\ 21,\ 21,\ 21.5,\ 22,\ 22,\ 24,\ 24,\ 25,\ 26,\ 27$$

a State the range of the temperatures.

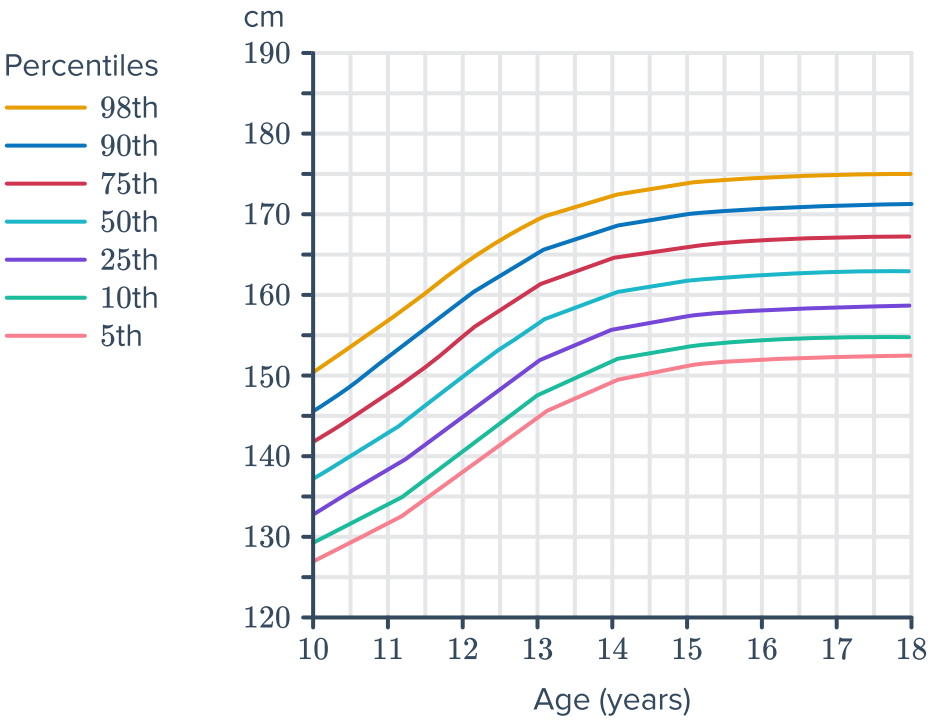
b Calculate the interquartile range of the temperatures.

c Find the population standard deviation, correct to one decimal place.

d Would the standard deviation or the interquartile range be the best measure of spread to support or counter the prediction of a huge variation in temperatures?

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31 Consider the following chart showing the range of heights for girls aged 10 to 18 years:



(Credit: State Government of Victoria, Department of Education and Training)
Rosey is aged 14 and is 160 cm tall. What percentage of girls her age are shorter than her?

32 The table shows the number of goals scored by a football team in each game of the year:

Score (x)	Frequency (f)
0	3
1	1
2	5
3	1
4	5
5	5

a In how many games were no goals scored?

b Determine the median number of goals scored.

c Calculate the mean number of goals scored each game, to two decimal places.

d Find the population standard deviation, correct to two decimal places.

e Find the variance, correct to three significant figures.

33 The scores obtained by two classes are given below:

- Red Class: $55, 57, 49, 58, 68, 57, 60, 53, 56, 51$
- Blue Class: $53, 57, 62, 51, 56, 62, 58, 55, 58, 51$

a Which class performed better on average? Use statistical calculations to justify your answer.

b Which class produced more consistent results? Use statistical calculations to justify your answer.

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34 Seven millionaires with an average net wealth of $\$41$ million with a standard deviation of $\$7$ million are having a party. Suddenly Carlos Slim, who has a net wealth estimated to be $\$31$ billion, walks into the room.

a Find the new average net wealth in the room. Give your answer rounded to the nearest million.

b Will the new standard deviation be higher, lower or unchanged from before?

c Will the new mode be higher, lower or unchanged from before if at least two of the original seven millionaires have the same net wealth?

d Will the range be higher, lower or unchanged from before?

35 Han, a cricketer, has achieved scores of 52, 20, 68, 70 and 150 in his first five innings this season. In his sixth innings, he scores 0.

a Describe how his season batting average changed from before to after the sixth inning.

b Describe how his standard deviation changed from before to after the sixth inning.

c Describe how his median score changed from before to after the sixth inning.

d Describe how his range changed from before to after the sixth inning.

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36 The table shows the heart rate data of a group of people after exercise:

Height of step	Stepping rate	Heart rate
Short step	Slow	89
Short step	Slow	91
Short step	Medium	106
Short step	Medium	105
Short step	Fast	124
Short step	Fast	128
Tall step	Slow	100
Tall step	Slow	96
Tall step	Medium	125
Tall step	Medium	129
Tall step	Fast	132
Tall step	Fast	127

a Complete the following table. Round all values to one decimal place.

Height of step	Data	Slow	Medium	Fast
Short step	Avg. heart rate	90.0		
	Standard deviation of heart rate	1.0		
Tall step	Avg. heart rate			
	Standard deviation of heart rate			

b Which of the combinations of step height and stepping rate generated the higher heart rate?

c Which combination of step height and stepping rate showed the least variability?

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